

POWER GENERATION WITH FOOTSTEPS OF STAIRS AND USES OF FOOTSTEPS FOR ADVANCE SECURITY PURPOSES

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1. Motivation

While humans walking, mechanical energy consists in footsteps totally wasting. This project is aimed to gain energy which is wasting while climbing or going down the stairs. The setup is Designed to convert Mechanical energy to Electrical energy. This generated energy is cost effective and nonhazardous for human. Another purpose of that project is identifying the person who climb or going down through the stairs. That is advantage when all security devices shut down since No power or malfunctioning.

2. Overview

In present word tends to generate Electricity in Environmentally sustainable ways. Still majority Electricity generated by Coal. Since combustion of coal and gasoline is hazardous for Environment and living being world tends to use renewable energies. In addition to renewable energy, Electricity can generate by humans' motions. Since every person has unique walking pattern, person identification system can be developed using walking pattern. Person identification feature can be used for Security purposes as well.

2.1 Significance of the project

The setup is designed for stairs while people climbing or going down through it. When a person going down, pressure applies on stair plate is more than person's weight. Because of additional potential energy gives additional force to footstep. That is beneficial than straight walkway. Since Person identification system combined to that project it gives significant function than other similar projects. The produce Electricity use for lightening of staircase and for person identification system without external power source.

There are related projects have done similar to that project. Those are mentioned below,

NTT Electricity Generating Shoe

Japanese telecommunications company NTT has developed shoes that generate electricity from walking. The shoes are designed to look like orthopedic sandals, and they can generate 1.2 watts of electricity from the weight placed on water cushioned soles. The water sloshes on mini turbines to generate electric power, which can power digital musical devices for as long as the wearer keeps walking.

Streetlights Powered by Sidewalks

Streetlights powered by sidewalks use special tiles or pavements that contain embedded generators that produce energy when stepped on. For example, Boulder Plaza in Las Vegas has special tiles with microgenerators that produce four to eight watts of energy with each step. The French city of Toulouse has energy-absorbing sidewalk panels that power streetlights.

Generate Power using Footsteps by a Spring Mechanism

This Mechanism consists of plate, spring, rods, Alternator. When foot compress the plate, plate moves downward and rods connected to plate goes downward, then there is Alternator joined to rod, since that translational motion generate Electricity.

Kinetic Dance Floors

Kinetic dance floors, also known as sustainable dance floors, use human movement to generate renewable energy. The floors are made of piezoelectric tiles that convert the kinetic energy of movement into electricity. For example, walking, dancing, or jumping on the floor can generate up to 35 watts of sustained output. The electricity can then be fed back into the floor to power lights, captured and stored, or reused to power an application.

2.2 Description of the project

The project aims to develop a system that utilizes the footsteps on stairs to generate electricity while concurrently employing advanced security features. The system will involve mechanical power generator embedded within each step to capture the kinetic energy produced by footsteps. Additionally, it will integrate security sensors such as motion detectors.

There is plate for each step and when person walking on plate, it will displace downward and the connecting rod connected to piston and step plate. Piston displace working fluid and using mini turbine and alternator Electricity produces.

Vibration sensor assembled under step plate gives more information about footsteps. From footprint details, who gone on step plate can be identify.

2.3 Background of the project

Electricity generation process cause pollution and consumed natural sources. In present world looking for renewable energy and environmental healthy electricity generations for decrease firing natural fuels. Small Electricity demand can be fulfilled by natural energy sources. Usually Humans kinetic energy wasting by walking, dancing and sum activities. Methods for extract that energies already developed in the world. This project will innovate power generator with person identify system.

3. Methodology

3.1 Design phase

For working fluid, we suggest two fluids, one is water another one is low viscos oil. We need to compare several types of fluid and choose best for working fluid. To analysis foot step pattern sensor need to be assembled. Therefor varies sensors need to be study such as Accelerometer, Eddy Current or Capacitive Displacement Sensors or specific tracker need to be designed.

3.2 Implementation phase

Designed the 3D model of reservoir turbine and pipes by 3D modelling software and test several fluid motions by software. Analysis the sensors data and make algorithm for identify persons.

3.3 Testing phase

The project completely simulates from simulation software.

3.4 Evaluation phase

Compare our project results with similar projects and decide what is effective. And also do feasibility study for cost effectiveness.

4. Features

Sustainable energy generation from human movement
Enhanced security through real-time monitoring and surveillance
Scalable and adaptable system suitable for various public spaces

5. Project Planning

TABLE 01- PROJECT PLANING

Task	Time allocated	Who engaged
Select working fluid	4 th week of semester	E/19/308
Select sensor or Design a equipment	5 th week	E/19/308 E/19/239
Power generator design with calculations.	6 th week	E/19/308
Calibrate sensor for application		E/19/239
Design The 3D model for setup	7 th week, 8 th week	E/19/308 E/19/239
Analyze the model with software. (Ansys)	9 th week	E/19/308
Analyze sensor data and make model to person identification	10 th week, 11 th week	E/19/239
Evaluate the project	12 th week	

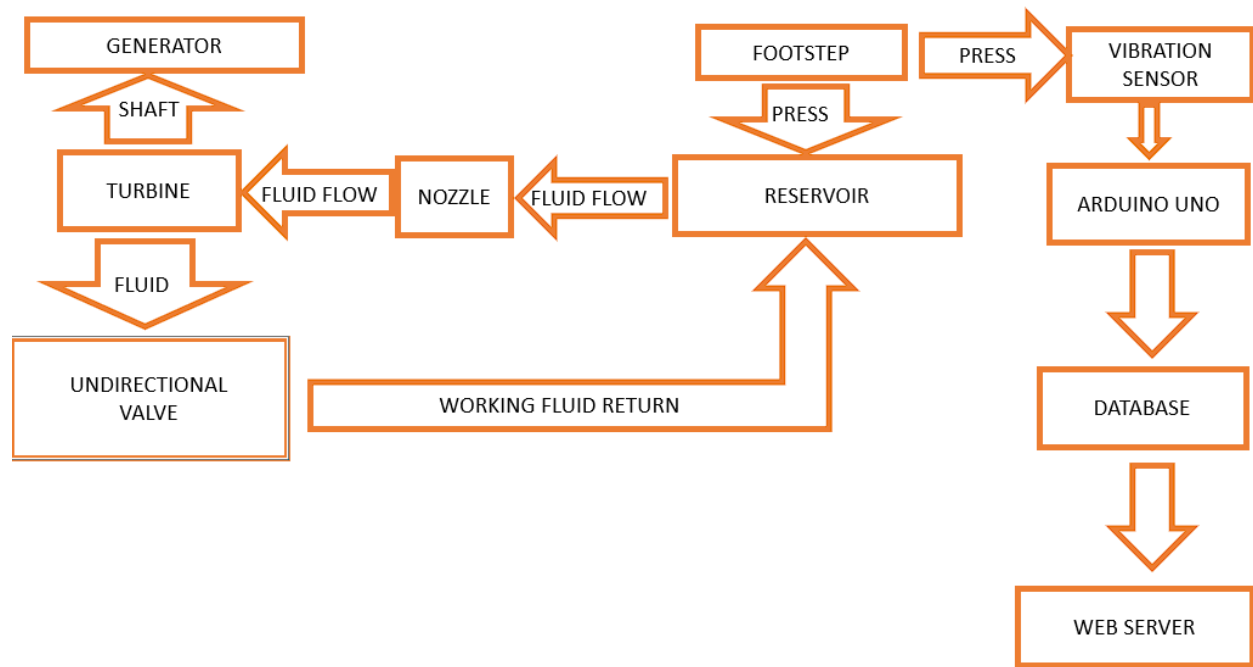
6. Hardware and Software requirement

Hardware -Turbine, Pipes, DC Generator, Vibration sensor, non-return valve, Arduino Uno

Software – Data Analysis Tools

7. Diagrammatic Representation

FIGURE 01 – FLOW DIAGRAM OF SETUP



8. References

1. <https://dl.acm.org/doi/10.1145/2699343.2699364>
2. <https://www.printedelectronicsworld.com/articles/1653/power-generating-shoes#:~:text=Depending%20on%20the%20speed%20of,to%20help%20wandering%20Alzheimer%20patients.&text=Japanese%20telecommunications%20company%20NTT%20claims,small%20devices%20on%20the%20go.>
3. <https://www.sciencedirect.com/science/article/abs/pii/S156625350600114X>